

TCP Connection Performance Analysis using Mininet and Wireshark

Assignment 2 Report

Name: Supratim Gogoi

Roll No: CS-BTC24-17

1. Objective

Implement a TCP client-server application in Mininet and compare Persistent TCP with New TCP Connection mode by measuring response time, throughput and verifying TCP behaviour using Wireshark.

2. Mininet Topology

The network consists of h1 (Server), s1 (Switch) and h2 (Client). The server runs on IP 10.0.0.1 using TCP port 5000, while the client uses IP 10.0.0.2.

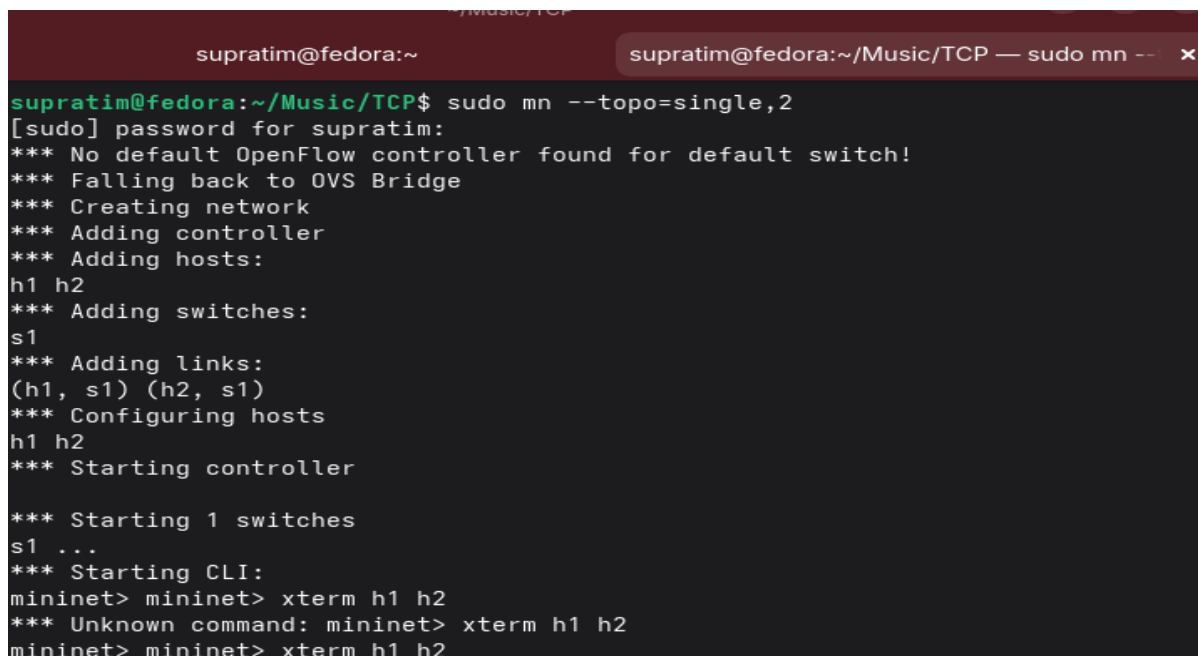
3. TCP Communication

The client sends messages in the format MSG_ID|MESSAGE_SIZE|MESSAGE_DATA. The server processes each message and returns ACK|MSG_ID|RECEIVED_SIZE. Persistent mode uses one connection for all messages, whereas New Connection mode establishes a separate TCP connection for every message.

4. Experimental Analysis

Persistent mode generally achieved lower response time because connection setup is performed only once. New Connection mode incurs additional delay due to repeated TCP handshakes and connection termination. Throughput increased with payload size in both modes.

Mininet Topology

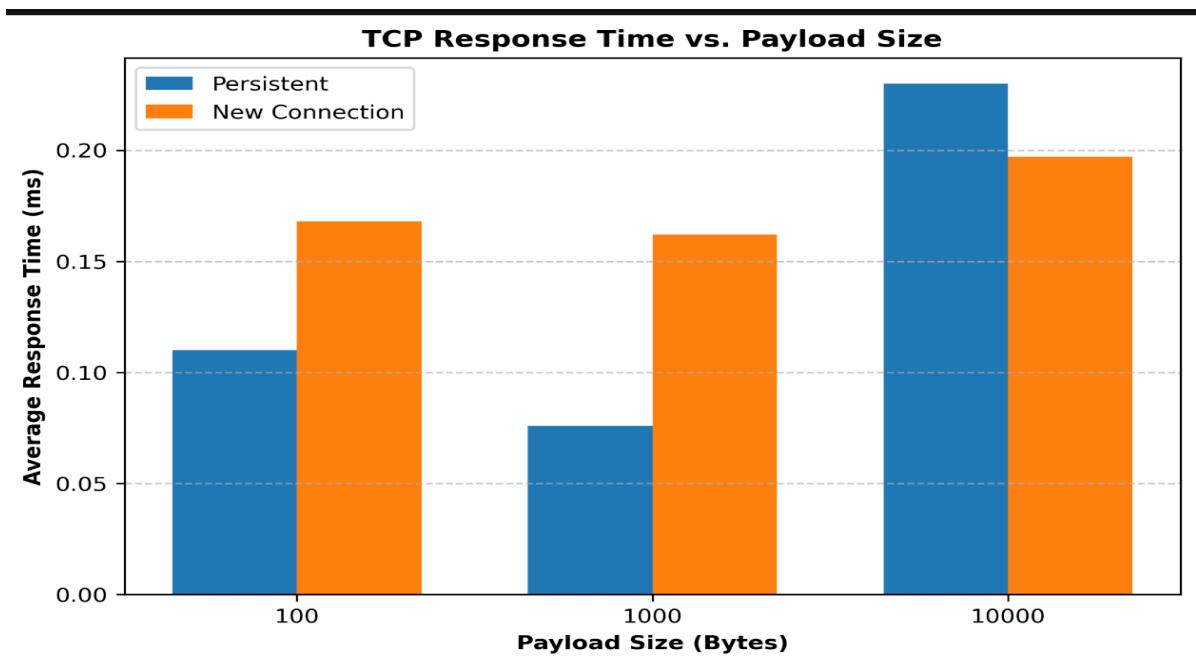


```
supratim@fedora:~  
supratim@fedora:~/Music/TCP — sudo mn --  
supratim@fedora:~/Music/TCP$ sudo mn --topo=single,2  
[sudo] password for supratim:  
*** No default OpenFlow controller found for default switch!  
*** Falling back to OVS Bridge  
*** Creating network  
*** Adding controller  
*** Adding hosts:  
h1 h2  
*** Adding switches:  
s1  
*** Adding links:  
(h1, s1) (h2, s1)  
*** Configuring hosts  
h1 h2  
*** Starting controller  
  
*** Starting 1 switches  
s1 ...  
*** Starting CLI:  
mininet> mininet> xterm h1 h2  
*** Unknown command: mininet> xterm h1 h2  
mininet> mininet> xterm h1 h2
```

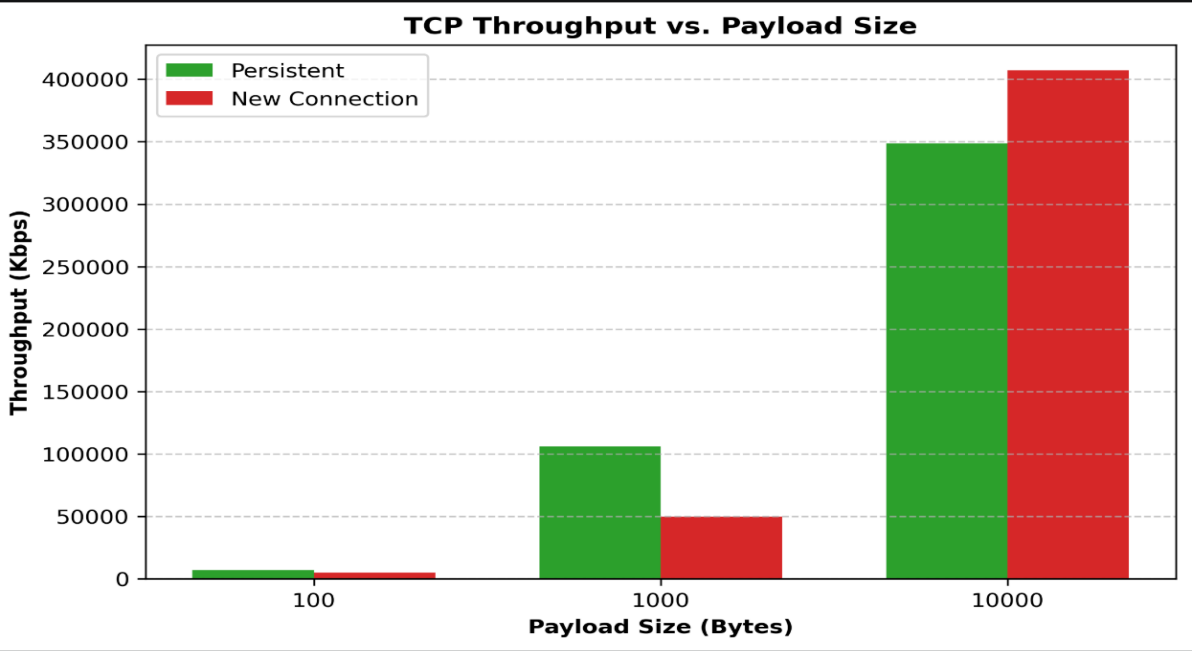
Client Output


```
result_table.csv X
result_table.csv
1 Mode,Payload_Size_Bytes,Total_Mess
2 persistent,100,10,0.11,7272.31
3 persistent,1000,10,0.076,105783.2
4 persistent,10000,10,0.23,348545.05
5 new_connection,100,10,0.168,4748.04
6 new_connection,1000,10,0.162,49526.84
7 new_connection,10000,10,0.197,407115.17
8
```

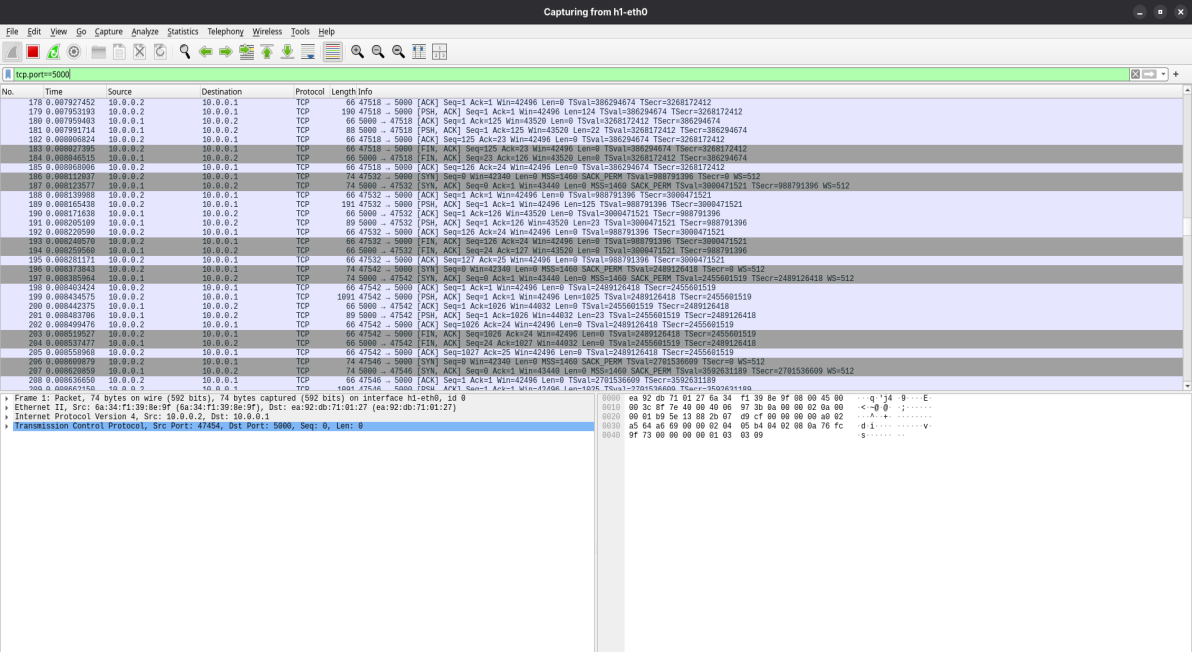
Response Time Graph



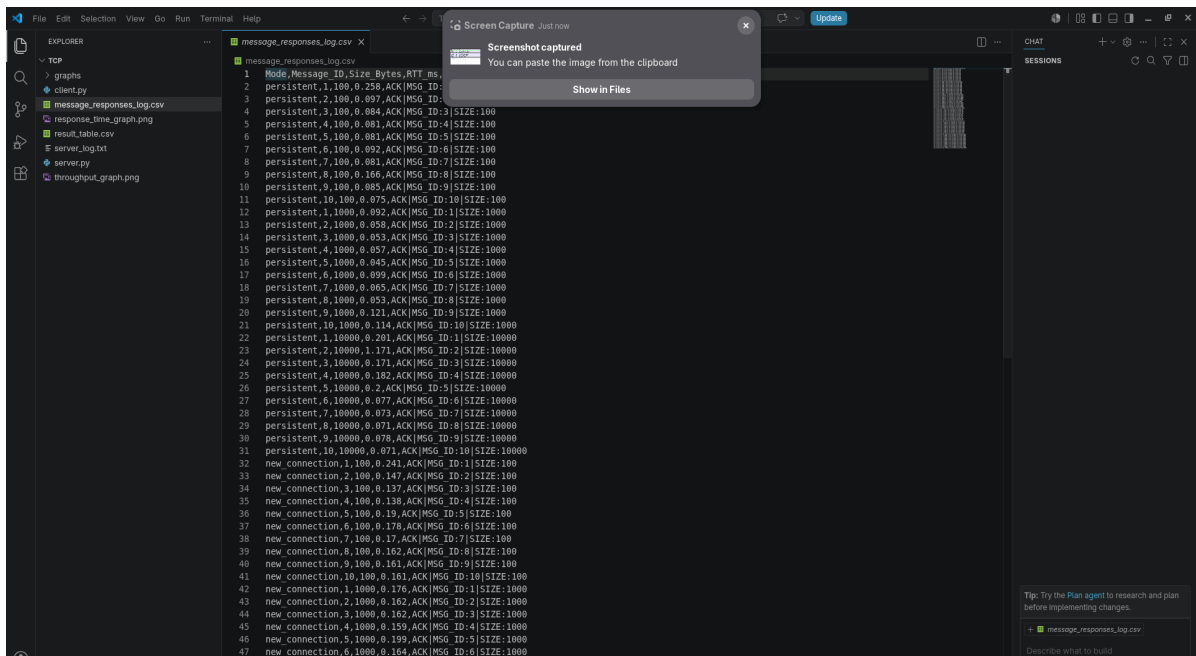
Throughput Graph



Wireshark Packet Capture



Message Response Log



Answers to Assignment Questions

Why does TCP use a three-way handshake?

It synchronizes sequence numbers and confirms both client and server are ready for reliable communication.

Which mode had lower response time?

Persistent mode showed lower response time for most payload sizes because it avoids repeated connection setup.

Why does new-connection mode have more overhead?

Every message requires a new SYN, SYN-ACK, ACK handshake and FIN sequence, increasing latency.

Which mode is better for CampusChat?

Persistent mode because it supports continuous communication with minimal delay.

Which mode is suitable for NetAttend?

New Connection mode because each attendance request is independent.

How does message size affect throughput?

Larger payloads generally improve throughput by reducing protocol overhead.

How did Wireshark help verify TCP behaviour?

Wireshark displayed TCP handshakes, ACK packets, data packets and FIN packets, confirming correct TCP operation.

Conclusion

The experiment successfully compared persistent and new TCP connection modes in Mininet. Persistent TCP provided better efficiency for repeated communication, while Wireshark confirmed correct TCP packet exchange and connection management.